A Incoherent Feedforward Loop (IFFL)

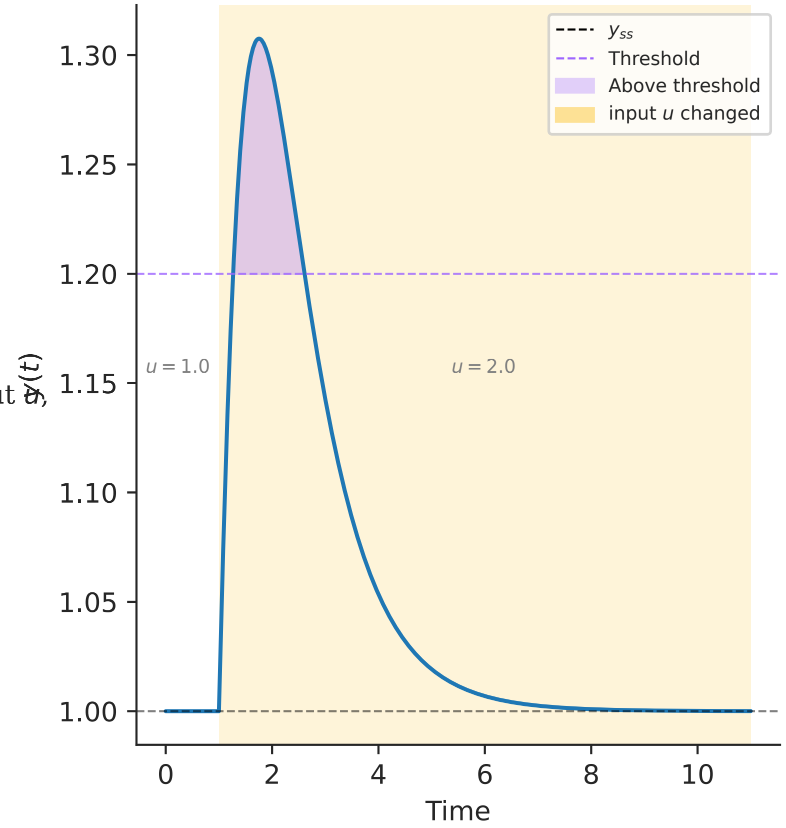
$$\frac{dx}{dt} = \alpha\, u - \delta\, x$$

$$\frac{dy}{dt} = \beta\, u - \gamma\, x\, y$$

Steady state: $y_{ss} = \frac{\beta\,\delta}{\gamma\,\alpha}$

The output's steady state is independent of the input u, making it robust to changes in u.

B y converges to y_{ss} after u changed



C Slow ramp then abrupt u jump